

Trial Sourcing under Uncertainty about Match Quality

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Abstract

This paper documents three facts about newly activated sourcing links in Colombian customs data. Relative to mature links, activated links start smaller, are more likely to be abandoned, and grow faster conditional on recording subsequent imports. Neither an adjustment-cost model nor a relationship-capital model reproduces all three facts when calibrated to the same four moments. I develop a trial sourcing model in which the first positive order reveals permanent source–product match quality. Uncertainty before sourcing leads firms to place smaller first orders and shifts selection from activation to continuation: low-quality matches are less likely to continue, while imports from high-quality matches grow. The model matches the four calibration moments and fits additional quarterly lifecycle moments more closely than the two alternatives. Relative to a full-information counterfactual, trial sourcing lowers import value from the initial cohort by 1.2 percent in the activation quarter but raises it by 9.6 percent by quarter four.

1 Introduction

New import links exhibit a striking lifecycle. In Colombian customs data, an activated origin begins at only 10.6 percent of mature-origin scale within the same importer–product–quarter. In the next quarter, 78.4 percent of activated origins record no imports. Four-quarter abandonment is 57.3 percent among activated links and 2.2 percent among mature links. Among activated links that record imports in quarter four, imports from activated origins grow by 0.252 log points, and the activated–mature conditional-growth gap is 0.318 log points. New links therefore start small, are frequently abandoned, and grow rapidly conditional on continuing to record imports.

Together, these facts present a puzzle. Heterogeneous-firm and sunk-cost models generate selection and persistence, but the first positive order has no distinct role (Das, Roberts and Tybout, 2007; Alessandria and Choi, 2014). Adjustment costs can generate small first orders and subsequent growth. When match quality is known before sourcing, however, poor matches are screened out before activation, leaving little early abandonment. Relationship capital can generate both small first orders and high early abandonment, but its gradual accumulation produces too much growth among activated links that continue to record imports.

I study this lifecycle using Colombian customs records from 2011 through 2019. The unit of observation is an importer–product–origin link, with products defined by harmonized HS6 categories. A link is activated when an importer begins sourcing a product from an origin after the link has recorded no imports for at least eight consecutive quarters. The importer must have sourced the same product from another origin during that period. The primary cohort contains

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606,871 activated links. I compare activated and mature origins within 127,975 importer–product–activation–quarter cells covering 7,133 firms. The comparison therefore holds the importer, product, and quarter fixed.

The three patterns hold across product groups. The initial-scale gap is negative in all 61 HS2 chapters that meet the sample requirement, and the four-quarter abandonment gap is positive in all 60 chapters for which it can be estimated. Among the 28 chapters with enough links to measure quarter-four growth, imports from activated origins grow in 27. Dropping any one chapter preserves the sign and at least half the magnitude of each baseline estimate.

I develop a *trial sourcing model* to explain this lifecycle. A monopolistically competitive importer chooses origins and orders within a product-level sourcing portfolio. Each source–product match has permanent high or low match quality. Match quality is unknown before sourcing and is revealed by the first positive order. Firm productivity is subject to an i.i.d. shock each period, and every positive order incurs a fixed order cost and a convex adjustment cost.

Match-quality revelation yields three implications. First, uncertainty makes the first order smaller than the order from a comparable mature origin known to be a high-quality match. Second, activated links are more likely to be abandoned because they include low-quality matches, whereas repeated importing selects mature links toward high-quality matches. Third, conditional on subsequent imports, imports from activated origins grow faster because their initial orders are smaller.

I calibrate the adjustment-cost, relationship-capital, and trial sourcing models to the same four moments: initial relative scale, the quarter-one no-import share, four-quarter abandonment, and the quarter-four conditional-growth gap. With match quality known before sourcing, the adjustment-cost model matches initial scale and conditional growth but sharply understates the quarter-one no-import share (10.2 versus 78.4 percent) and four-quarter abandonment (2.2 versus 57.3 percent). The relationship-capital model better matches these margins but overpredicts the conditional-growth gap (0.403 versus 0.318 log points). The trial sourcing model matches all four moments.

I then compare the models using quarterly lifecycle moments that are not calibration targets. At age eight, the no-import share is 47.3 percent in the data, compared with 38.9 percent under trial sourcing, 1.6 percent under adjustment costs, and 31.2 percent under relationship capital. Conditional on imports, the activated–mature import-value ratio rises from 0.128 to 0.168 in the data, compared with 0.116 to 0.192 under adjustment costs and trial sourcing and 0.123 to 0.230 under relationship capital. Trial sourcing also gives the closest fit when links that record no imports are included.

Finally, I compare trial sourcing with a full-information counterfactual that reveals match quality before activation while holding the parameter vector, adjustment costs, and distribution of potential matches fixed. Trial sourcing activates more links but places smaller first orders. The two margins nearly offset in the activation quarter, when import value from the initial cohort is 1.2 percent lower than under full information. By quarter four, it is 9.6 percent higher. Because firms cannot screen on match quality before activation under trial sourcing, low-quality matches make up a larger share of activated links.

This paper makes three contributions. First, it documents the joint lifecycle of activated origins within existing importer–product portfolios rather than at the firm or destination level. Second, it introduces match-quality revelation through the first positive order into a dynamic global sourcing model and derives its implications for initial scale, subsequent imports, and conditional growth. Third, it compares trial sourcing with adjustment costs and relationship capital by calibrating all three models to the same four moments and evaluating their fit to additional quarterly moments.

1.1 Related literature

This paper relates to work on post-entry trade dynamics, learning through trade, relationship capital, global sourcing, and the measurement of trade-relationship duration. [Alessandria, Arkolakis and Ruhl \(2021\)](#) and [Bernard and Moxnes \(2018\)](#) survey the broader literatures on firm dynamics in trade and firm-to-firm trade networks.

Entry and post-entry dynamics. Canonical heterogeneous-firm models link trade participation to firm productivity and trade costs ([Melitz, 2003](#)). Dynamic models introduce sunk entry costs and persistent shocks to explain persistence and exit ([Das, Roberts and Tybout, 2007](#); [Alessandria and Choi, 2014](#)), but the first positive order plays no informational role in these models.

Models of post-entry trade growth generate richer dynamics. In [Ruhl and Willis \(2017\)](#), foreign demand rises with export tenure, producing small initial exports and gradual growth, while stochastic entry costs allow less productive plants to enter and exit early. [Piveteau \(2021\)](#) obtains similar dynamics through the accumulation of foreign consumers. On the importing side, [Gimenez-Perales \(2026\)](#) uses tenure-dependent fixed costs to generate the gradual rise in new importers’ conditional survival rates and import shares toward the levels of established importers.

These models describe firm- or market-level lifecycles. Related empirical work documents short-lived trade relationships and small initial purchases at the product–country level. [Besedeš and Prusa \(2006\)](#) show that many U.S. import relationships are short-lived. [Besedeš \(2008\)](#) further shows that many begin with small import values and relates their initial scale and duration to search costs. This paper instead compares activated and mature origins within the same importer–product–quarter.

Learning and experimentation through trade. [Rauch and Watson \(2003\)](#) provide the closest theoretical antecedent. In their model, a matched buyer and supplier may begin with a small trial order to learn whether the supplier can carry out a larger project. [Araujo, Mion and Ornelas \(2016\)](#) study how contract enforcement and export experience shape firm export dynamics in an environment with incomplete information. Stronger contract enforcement leads to larger initial exports and longer export relationships but slower growth among continuing exporters.

At the exporter–market level, [Nguyen \(2012\)](#) models demand uncertainty revealed after entry, while [Albornoz et al. \(2012\)](#) show how information from early export experience guides sequential entry across destinations. [Timoshenko \(2015\)](#) uses the age dependence of sales to distinguish learning from sunk-cost state dependence. [Eaton et al. \(2025\)](#) combine customer search, visibility effects, and learning about product appeal. [Fanelli, Hallak and Yin \(2026\)](#) propose that firms experiment to resolve an uncertain component in foreign-market profitability, to explain the different survival rates of first-time entrants and re-entrants in export markets.

This paper studies experimentation within the importer’s product-level origin portfolio. The observed link is an importer–product–origin rather than a named buyer–supplier relationship, and the first positive order reveals permanent source–product match quality.

Relationship capital. Several models generate gradual trade growth through an accumulating relationship-specific state. [Drozd and Nosal \(2012\)](#) model foreign customers as capital built through marketing. [Macchiavello and Morjaria \(2015\)](#) show how reputation and relational contracting make a trading relationship more valuable with age. [Monarch and Schmidt-Eisenlohr \(2023\)](#) document that traded quantities and survival probabilities rise with relationship age and model these patterns as the accumulation of relationship capital. These studies motivate the relationship-capital benchmark considered here. Relationship capital raises the value of a link gradually, whereas trial sourcing changes the importer’s information after the first positive order.

Global sourcing portfolios. The model builds on global sourcing frameworks in which the value of one origin depends on the importer’s other origins. [Kasahara and Lapham \(2013\)](#) jointly model firms’ importing and exporting decisions. [Antràs, Fort and Tintelnot \(2017\)](#) show that source-country choices interact through the firm’s cost function, so the return to adding an origin depends on the rest of the sourcing set. [Gimenez-Perales \(2024\)](#) models costly expansion of importers’ potential supplier sets and shows how input substitutability links decisions across suppliers. [Hoang \(forthcoming\)](#) combines this cross-country complementarity with importing costs that make new supplier relationships costlier than existing ones. This paper adds uncertainty about source–product match quality to this portfolio problem.

Shipment timing and temporal aggregation. Temporary gaps between shipments need not mark the end of a sourcing link. Fixed shipment costs and inventory management generate infrequent and lumpy trade ([Alessandria, Kaboski and Midrigan, 2010](#); [Kropf and Sauré, 2014](#)). Entry partway through a calendar period can also depress measured initial trade and inflate subsequent growth ([Bernard et al., 2017](#)). I therefore require eight preceding quarters with no imports to define link activation, define abandonment as four consecutive post-activation quarters with no imports, and check alternative windows and annual aggregation.

Section 2 describes the data, defines the lifecycle measures, and presents the three empirical facts. Section 3 develops the adjustment-cost, relationship-capital, and trial sourcing models. Section 4 presents the calibration, model comparison, and counterfactual results. Section 5 concludes.

2 Empirical results

2.1 Data and link definition

Customs data and product harmonization I use monthly Colombian customs declarations from 2011 to 2019.¹ Each record reports the importer, HS6 product, origin country, import value, physical units, and date. I harmonize HS product codes across the 2007, 2012, and 2017 revisions into 5,229 stable product categories. Some HS6 codes cannot be matched one-to-one across revisions because they were split or merged over time. I therefore combine these codes into the smallest product groups that remain comparable across years. The resulting classification covers 99.1% of total import value. Finally, I aggregate the data to the importer–product–origin–quarter level, which yields 8.88 million observations.

Because Colombia introduced the compound tariffs at issue in WTO dispute DS461 during the sample period, the baseline sample excludes all HS2 chapters containing directly affected products ([World Trade Organization, 2013](#)). In an alternative sample, I exclude only the directly treated HS6 products and retain unaffected products from the same chapters.²

Link activation Let v_{fhot} denote firm f ’s import value for product h from origin o in quarter t . I define a *link activation* for importer–product–origin link (f, h, o) in quarter t by three conditions.

¹I initially restrict the analysis to 2011–2019 as a benchmark to establish the trial-sourcing lifecycle before 2020 – 2021 pandemic, because I would like to remove the pandemic disruptions to shipping, demand, and supplier availability. I will next extend the panel through 2025: retain 2011–2019 for calibration, report 2020–2021 separately as a pandemic stress test, and use 2022–2025 for out-of-period validation by recomputing the same lifecycle moments without changing the model.

²Colombia introduced the tariffs examined in DS461 in 2013, within the sample period. Directly affected products account for only 1.8% of total Colombian import value, so the remaining sample still covers most Colombian import value.

First, firm f records no imports of product h from origin o during the preceding eight quarters.³ Second, the firm imports the same product from at least one other origin during this period.⁴ Third, the firm records imports of product h from origin o in quarter t .

I call quarter t the *activation quarter*, link (f, h, o) an *activated link*, and origin o an *activated origin*. The baseline sample contains 606,871 activated links. Of these, 476,733 are observed for the first time in the panel, while 130,138 are reactivations following at least eight consecutive quarters that record no imports.

Link maturity For each activated link (f, h, o) , an origin o' serving the same importer f and product h meets the *maturity criterion* if it records imports in the activation quarter and in each of the four preceding quarters. I call o' a *mature origin* and its importer–product–origin relationship a *mature link*. This criterion identifies 230,366 mature links.

The *activated–mature comparison* consists of 127,975 importer–product–activation–quarter cells across 7,133 firms, each containing at least one activated origin and one mature origin. Within each cell, I calculate the mean outcome separately for activated and mature origins and take the difference between the two. I call each difference an *activated–mature gap*. I then average these cell-level gaps across the sample, giving every cell equal weight regardless of how many origins it contains.

Link abandonment I classify an activated link as abandoned if it records no imports in all four consecutive quarters after the activation quarter. A link that records no imports in the next quarter but returns within this four-quarter window is not classified as abandoned.

Abandonment is much more common among activated links than among mature links in the same importer–product–activation–quarter cells. Over the four consecutive quarters after activation, 57.3 percent of activated links are abandoned, compared with only 2.2 percent of mature links, an activated–mature abandonment gap of 55.1 percentage points.

Appendix A describes product harmonization, sample censoring, and the alternative activation and sample definitions used in the robustness checks.

2.2 Lifecycle measures

Let $c = (f, h, t)$ index an importer–product–activation–quarter cell. For any link-level outcome y_ℓ , let $\mathcal{L}_c^A(y)$ and $\mathcal{L}_c^M(y)$ contain the activated and mature links in cell c for which the outcome is observed. For group $r \in \{A, M\}$, define the cell mean as

$$\bar{y}_c^r = \frac{1}{|\mathcal{L}_c^r(y)|} \sum_{\ell \in \mathcal{L}_c^r(y)} y_\ell, \quad r \in \{A, M\}.$$

The corresponding activated–mature gap is

$$\Delta(y) = \frac{1}{|\mathcal{C}(y)|} \sum_{c \in \mathcal{C}(y)} (\bar{y}_c^A - \bar{y}_c^M),$$

³The eight-quarter no-import window reduces the risk of classifying infrequent and lumpy importing as a link activation. Delivery lags, transaction-level economies of scale, inventory management, and fixed costs per shipment can lead firms to place large orders at low frequencies, generating temporary zeros in customs records (Alessandria, Kaboski and Midrigan, 2010; Kropf and Sauré, 2014).

⁴This restriction ensures that link activation captures the addition of origin o to an existing product-level sourcing portfolio, rather than the firm’s entry or re-entry into importing product h . Without this restriction, the link-activation sample would combine product entry, product re-entry, and origin diversification. The restriction therefore isolates origin diversification within an existing importer–product portfolio.

where $\mathcal{C}(y)$ contains cells with both activated and mature outcomes. Thus, links receive equal weight within each role and each cell receives equal weight in the aggregate.

Initial scale The primary initial-scale measure is $\Delta(\log v_{\ell t})$, where $v_{\ell t}$ is the link’s import value in the activation quarter. I exponentiate this gap to report activated-origin scale relative to mature-origin scale under equal cell weighting.

I also report two link-level measures: the activated origin’s share of importer–product import value in the activation quarter and its import value in that quarter relative to the importer’s average quarterly import value for the product over the preceding eight quarters.

Abandonment The next-quarter no-import indicator is

$$D_{\ell 1}^0 = \mathbf{1}\{v_{\ell, t+1} = 0\}.$$

The abandonment indicator is

$$D_{\ell}^{\text{abd}} = \mathbf{1}\left\{\sum_{j=1}^4 v_{\ell, t+j} = 0\right\}.$$

I report both measures separately for activated and mature links, together with the corresponding activated–mature gaps. Each measure includes only links observed through its full horizon.

Post-activation growth For a link that records imports at horizon k , conditional log growth since activation is⁵

$$G_{\ell k} = \log v_{\ell, t+k} - \log v_{\ell t}, \quad v_{\ell, t+k} > 0, \quad k = 1, \dots, 4.$$

I report conditional log growth since activation for activated and mature links and the corresponding gap $\Delta(G_{\ell k})$. Because this measure conditions on imports at horizon k , I also report zero-inclusive growth.⁶

2.3 The trial sourcing lifecycle

Activated links display three clear patterns: they start small, often record no imports soon after activation, and expand rapidly when they continue to record imports.

Fact 1: Activated links start small Within the same importer–product–quarter, the mean activated–mature log-value gap is -2.242 . Exponentiating this gap implies that activated origins begin at 10.6% of mature-origin scale in geometric-mean terms. The median activated origin’s share of importer–product import value is 4.1% in the activation quarter.

Figure 1 shows that the initial-scale gap is pervasive across importer–product–quarter cells. Panel A shows that 85.2% of the cell-level activated–mature log-value gaps are negative. In Panel B, the median import value is \$725 for activated origins and \$7,862 for mature origins. Panel C shows that the gap remains large across alternative product definitions, firm samples, no-import windows, and annual aggregation; activated origins begin at 9.8% to 13.7% of mature-origin scale.

⁵I define post-activation growth as the cumulative growth since activation, directly comparing import value in quarter k with import value in the activation quarter. It has not been normalized to a per-quarter rate.

⁶Let $\tilde{v}_{ht}^M$ denote the median import value among mature links for product h in quarter t . Zero-inclusive growth since activation is $Z_{\ell k} = \text{asinh}(v_{\ell, t+k}/\tilde{v}_{ht}^M) - \text{asinh}(v_{\ell t}/\tilde{v}_{ht}^M)$, where $v_{\ell, t+k} = 0$ when the link records no imports.

Table A.1 reports the estimates and confidence intervals. Panel D reports two link-level measures of scale: the activated origin’s share of importer–product import value in the activation quarter and its import value in that quarter relative to the importer’s average quarterly import value for the product over the preceding eight quarters. At the 25th, 50th, and 75th percentiles, the share of importer–product import value is 0.7%, 4.1%, and 16.9% for activated origins, while the corresponding ratios to pre-activation import value are 0.7%, 4.1%, and 18.2%.

Fact 2: Early abandonment is high In the quarter after link activation, 78.4% of activated origins record no imports, compared with 10.2% of mature origins. Over the four consecutive quarters after activation, 57.3% of activated links record no imports in every quarter and are therefore classified as abandoned, compared with 2.2% of mature links.

Figure 2 documents post-activation imports and abandonment. Panel A plots the fraction of links that record imports within k quarters after activation. Activated links are consistently less likely than mature links to record subsequent imports. The activated–mature gap is 68.2 percentage points at $k = 1$ (21.6 versus 89.8 percent) and remains 45.7 percentage points at $k = 8$ (52.7 versus 98.4 percent). Panels B–D rematch mature origins within the same importer–product–quarter at each age. Panel B restricts the risk set to links that record imports in every quarter through age a . The activated–mature gap in the probability of recording no imports next quarter falls from 68.2 percentage points at age zero to 17.1 percentage points at age four. Panel C plots the probability that a link records no imports in all four consecutive quarters after the current age; at age zero, this outcome is abandonment. The corresponding gap falls from 55.1 percentage points at age zero to 6.2 percentage points at age four. Panel D requires a link to record imports only at the current age, allowing it to return after an earlier quarter with no imports. With this definition, the activated–mature gap in the probability of recording no imports next quarter remains 41.4 percentage points at age four.

Fact 3: Activated links with subsequent imports expand Among links that record imports four quarters after link activation, imports from activated origins grow by 0.252 log points from the activation quarter. Imports from mature origins fall by 0.118 log points. Within matched importer–product–quarter cells, the activated–mature growth gap is 0.318 log points.

Figure 3 traces these dynamics across the four post-activation quarters. Panel A shows that imports from activated origins grow by 0.136 log points at $k = 1$ and 0.254 at $k = 3$, then remain near that level at $k = 4$. Imports from mature origins decline throughout.

Panels B and C show how selective this growth is. By $k = 4$, 42.7% of activated links have recorded imports in at least one post-activation quarter. Among activated links that record imports at $k = 4$, imports grow by 28.7% since activation. When links that record no imports at $k = 4$ are included as zeros, however, zero-inclusive growth since activation is -0.299 . This contrast reflects survivor selection: expansion is concentrated among activated links that continue to record imports, whereas the zero-inclusive measure is negative because many activated links record no imports in quarter four.

Panel D shows how imports from incumbent origins and total importer–product imports evolve after link activation. At $k = 4$, the median change in imports from other origins is $-\$677$, while the median change in total importer–product imports is $-\$1,515$. Both outcomes therefore decline after activation.

Robustness across products and cohorts The three lifecycle patterns hold across products and cohorts. Among HS2 chapters with sufficient links and firms, activated links start smaller

than mature links in all 61 chapters, and their four-quarter abandonment rate is higher in all 60 chapters. Among the 28 chapters with enough links that record imports four quarters after activation, imports from activated origins grow in 27, with statistically significant growth in 14. Moreover, no single HS2 chapter drives the aggregate results. Removing any one chapter preserves the sign and at least half the magnitude of every estimate. Appendix B reports the full results.

3 A model of trial sourcing

This section develops three models. The adjustment-cost and relationship-capital models assume that permanent source–product match quality is known before sourcing. Their dynamics arise from gradual order adjustment or the accumulation of relationship capital. The trial sourcing model instead assumes that permanent match quality is unknown before sourcing and revealed by the first positive order. I derive the implications of this information structure for the three lifecycle facts in Section 2.3.

All three models share the same sourcing-portfolio structure. As in [Antràs, Fort and Tintelnot \(2017\)](#), the firm jointly chooses which origins to source from and how much to order from each. Because inputs from different origins enter a common imported-input aggregator, the return to ordering from one origin depends on orders from the others.

3.1 Alternative dynamic models

Both models assume that the importer observes source–product match quality before sourcing. The first emphasizes the dynamics generated by convex adjustment costs. The second adds relationship capital as an additional source of age dependence.

The adjustment-cost model. The adjustment-cost model generates order dynamics through convex costs of changing origin-specific imports. Let s denote current productivity and delivered import prices, and let $\mathbf{x}_-$ denote the previous-period order vector. The firm observes the vector of permanent match efficiencies $\boldsymbol{\lambda}$ and chooses current orders $\mathbf{x}$. Origin o belongs to the current sourcing set when $x_o > 0$. The recursive problem is

$$V(\boldsymbol{\lambda}, \mathbf{x}_-, s) = \max_{\mathbf{x} \in \mathcal{X}(\boldsymbol{\lambda}, \mathbf{x}_-, s)} \left\{ \Pi(\mathbf{x}; s, \boldsymbol{\lambda}) - \sum_o \mathbf{1}\{x_o > 0\} \kappa - \sum_o \Psi_o(x_o, x_{o,-}; p_o^m) + \beta \mathbb{E} V(\boldsymbol{\lambda}, \mathbf{x}, s') \right\},$$

where Π is variable profit, κ is the fixed cost of placing a positive order, Ψ_o is the cost of adjusting the order from its previous-period level, and $\mathcal{X}$ is the set of order vectors that satisfy the positive-order condition defined below. As in the trial sourcing model, every positive order must generate enough current operating surplus to cover the fixed order cost, evaluated here at the observed $\boldsymbol{\lambda}$.

Initial scale. Because match quality is known before activation, the model contains no informational motive for a small first order. Convex adjustment costs can nevertheless make the first order small because a newly activated origin has no previous-period order.

Abandonment. Because match quality is known before sourcing, the first positive order does not change the firm’s information. Productivity and delivered-price shocks, selection, and differences in previous-period orders can still generate periods with no imports, but the calibrated model produces too little early abandonment.

Conditional growth. Convex adjustment costs can generate faster growth among activated links that continue to record imports as their initial order gap closes.

The model can therefore account for small initial orders and conditional growth, but it generates far less early abandonment than observed in the data.

The relationship-capital model. This model introduces a relationship-specific stock, h_{ot} , that affects profit. Its law of motion is

$$h_{o,t+1} = (1 - \delta_h)h_{ot} + g(i_{ot}), \quad \delta_h \in [0, 1], \quad g'(i) > 0,$$

with depreciation rate δ_h and investment function $g(\cdot)$. This specification captures the customer-capital and market-penetration mechanisms used to generate gradual trade adjustment (Drozd and Nosal, 2012). It also captures the age-dependent relationship value documented in matched importer–exporter data (Monarch and Schmidt-Eisenlohr, 2023).

Initial scale. A newly activated link begins with little relationship capital, which can make its first order small. Continued sourcing builds relationship capital and raises the value of future orders.

Abandonment. Based on the calibration in Section 4, the model predicts that 85.3% of activated links record no imports in the next quarter, compared with 78.4% in the data. It predicts four-quarter abandonment of 52.7%, compared with 57.3% in the data.

Conditional growth. Accumulating relationship capital can generate gradual growth among links that continue to record imports. In the common calibration, the model predicts a quarter-four conditional-growth gap of 0.403 log points, exceeding the observed gap of 0.318 log points.

The relationship-capital model therefore comes closer to the observed no-import rates than the adjustment-cost model, but it overpredicts next-quarter no imports and conditional growth and does not match the three lifecycle facts jointly.

3.2 The trial sourcing model under uncertainty about match quality

Demand, technology, and profits. The firm produces a differentiated final good under monopolistic competition. I suppress firm and product indices and use o for origins and t for time. The final demand is

$$y_t = \mathcal{D}p_t^{-\sigma}, \quad \sigma > 1,$$

where p_t is the output price. Production uses a CES composite of imported inputs:

$$y_t = \varphi_t M(\mathbf{x}_t; \boldsymbol{\lambda}), \quad \varphi_t = \bar{\varphi} \varepsilon_t, \quad \log \varepsilon_t \stackrel{iid}{\sim} F_\varepsilon,$$

$$M(\mathbf{x}_t; \boldsymbol{\lambda}) = \left[\sum_{o \in \mathcal{O}} (\lambda_o x_{ot})^\rho \right]^{1/\rho}, \quad \rho = \frac{\eta - 1}{\eta} \in (0, 1),$$

where x_{ot} is the order quantity from origin o , $\eta > 1$ is the elasticity of substitution across origins, and λ_o is source–product match efficiency. The firm observes the transitory productivity shock ε_t before ordering.

Let p_{ot}^m be the delivered import price and $v_{ot} = p_{ot}^m x_{ot}$ the resulting import value. Delivered import prices follow an exogenous Markov process. With $\alpha = (\sigma - 1)/\sigma$, sales and variable profit are

$$\mathcal{R}(\mathbf{x}_t; \varphi_t, \boldsymbol{\lambda}) = \mathcal{D}^{1/\sigma} [\varphi_t M(\mathbf{x}_t; \boldsymbol{\lambda})]^\alpha, \tag{1}$$

$$\Pi(\mathbf{x}_t; \varphi_t, \mathbf{p}_t^m, \boldsymbol{\lambda}) = \mathcal{R}(\mathbf{x}_t; \varphi_t, \boldsymbol{\lambda}) - \sum_o p_{ot}^m x_{ot}. \tag{2}$$

For a sourcing set $\mathcal{S}$ and known match efficiencies, the imported-input price index is

$$c(\mathcal{S}; \mathbf{p}_t^m, \boldsymbol{\lambda}) = \left[\sum_{o \in \mathcal{S}} \left(\frac{p_{ot}^m}{\lambda_o} \right)^{1-\eta} \right]^{1/(1-\eta)}.$$

The optimal output price, sales, and variable profit are

$$p_t^* = \frac{\sigma}{\sigma-1} \frac{c}{\varphi_t}, \quad \mathcal{R}_t^* = \mathcal{D} \left(\frac{\sigma}{\sigma-1} \frac{c}{\varphi_t} \right)^{1-\sigma}, \quad \Pi_t^* = \frac{\mathcal{R}_t^*}{\sigma}. \quad (3)$$

Holding orders from all other origins fixed, define the operating surplus generated by an order of x from origin o as

$$\pi_o(x; \mathbf{x}_{-o}, s, \lambda) = \Pi(x, \mathbf{x}_{-o}; s, \lambda) - \Pi(0, \mathbf{x}_{-o}; s), \quad s = (\varphi, \mathbf{p}^m).$$

Uncertainty about match quality and timing. Each source–product match has permanent high or low match quality. In the model, match quality is represented by the binary match-efficiency parameter

$$\lambda_o \in \{\lambda_L, \lambda_H\}, \quad 0 < \lambda_L < \lambda_H, \quad \Pr(\lambda_o = \lambda_H) = \mu \in (0, 1).$$

λ_H corresponds to the high-quality type and λ_L to the low-quality type. The information state for origin o is $I_{ot} \in \{U, L, H\}$. Before the first positive order, $I_{ot} = U$ and match quality is unknown. The first positive order reveals whether the match is low or high quality, so the next-period information state becomes L or H . This information is permanent: once $I_{ot} \in \{L, H\}$, it remains unchanged during periods with no order and after re-entry. Formally,

$$I_{o,t+1} = \begin{cases} U, & I_{ot} = U \text{ and } x_{ot} = 0, \\ L, & I_{ot} = U, x_{ot} > 0, \text{ and } \lambda_o = \lambda_L, \\ H, & I_{ot} = U, x_{ot} > 0, \text{ and } \lambda_o = \lambda_H, \\ I_{ot}, & I_{ot} \in \{L, H\}. \end{cases}$$

Fixed and adjustment costs. Every positive order from an origin in a period incurs the same fixed cost $\kappa > 0$. Given information state I_o , the expected operating surplus generated by ordering x from origin o is $\bar{\pi}_o(x; \mathbf{x}_{-o}, s, I_o)$. A positive order must generate enough current operating surplus to cover the fixed cost:

$$\begin{aligned} \mathcal{X}_o^+(\mathbf{x}_{-o}, s, I_o) &= \{x > 0 : \bar{\pi}_o(x; \mathbf{x}_{-o}, s, I_o) \geq \kappa\}, \\ \underline{x}_o(\mathbf{x}_{-o}, s, I_o) &= \inf \mathcal{X}_o^+(\mathbf{x}_{-o}, s, I_o). \end{aligned}$$

Whenever this set is nonempty, continuity and $\kappa > 0$ imply a strictly positive minimum order size, $\underline{x}_o > 0$. The firm therefore cannot reveal match quality through an arbitrarily small order.

The common adjustment cost is

$$\Psi_o(x, x_-; p_o^m) = \frac{\gamma p_o^m}{2} (x - x_-)^2, \quad \gamma \geq 0.$$

Recursive problem. Let $\mathbf{I}$ collect the information states across origins, and let $\mathcal{X}(\mathbf{I}, \mathbf{x}_-, s)$ impose this positive-order condition jointly across origins. The firm's value is

$$V(\mathbf{I}, \mathbf{x}_-, s) = \max_{\mathbf{x} \in \mathcal{X}(\mathbf{I}, \mathbf{x}_-, s)} \left\{ \mathbb{E}[\Pi(\mathbf{x}; s, \boldsymbol{\lambda}) \mid \mathbf{I}] - \sum_o \mathbf{1}\{x_o > 0\} \kappa - \sum_o \Psi_o(x_o, x_{o,-}; p_o^m) + \beta \mathbb{E}V(\mathbf{I}', \mathbf{x}, s') \right\}.$$

Solution. I solve the model in recursive partial equilibrium, taking final demand $\mathcal{D}$ and the exogenous state $s = (\varphi, \mathbf{p}^m)$ as given. A solution consists of the value function, the order policy, the output-pricing rule, and the transition of the information state.

3.3 Lifecycle implications

For the comparative statics below, I hold fixed the orders and policy rules for all other origins. Let $V_o(x_-, s; j)$ denote the continuation value for origin o after match type $j \in \{L, H\}$ is revealed. Appendix C provides its recursive representation and proves the results.

The first result concerns initial scale. Let $x_o^U(s)$ denote an optimal positive trial order, and let $x_o^H(x_-, s)$ denote the optimal positive order from a revealed high-quality match. An otherwise comparable mature link enters with a positive previous-period order, $x_- > 0$.

Proposition 1 (Initial-scale gap). *Suppose the objective conditional on a positive order is strictly concave, current operating surplus has strict increasing differences in order size and match efficiency, and $V_{o,x_-}(\cdot; H) \geq V_{o,x_-}(\cdot; L)$. Holding productivity, delivered import prices, and orders from other origins fixed, any interior trial order satisfies*

$$x_o^U(s) < x_o^H(x_-, s).$$

At a common delivered import price, the same ranking holds for import values.

Two forces make the first order smaller. First, the prior averages the marginal return across high- and low-quality matches, below the marginal return for a known high-quality match. Second, a mature high-quality match faces a lower adjustment cost because it has a positive previous-period order.

The second result concerns subsequent imports.

Proposition 2 (Next-quarter no-import gap). *Suppose $0 < \mu < 1$ and productivity has positive density between the two type-specific sourcing cutoffs. Holding delivered import prices and orders from other origins fixed:*

(i) No adjustment cost. *If $\gamma = 0$, then*

$$\Pr(x_{o,t+1} = 0 \mid \text{link activation at } t) > \Pr(x_{o,t+1} = 0 \mid \text{maturity criterion at } t).$$

(ii) Positive adjustment cost. *If $\gamma > 0$, assume: (a) revealed-type order and participation policies are weakly increasing in match efficiency and the previous-period order; (b) at each relevant history, the high-quality type places a positive order over a strictly larger set of productivity states; and (c) in a common state, a mature link of either type places an order no smaller than the trial order. Then the same inequality holds.*

The first positive order reveals match quality. High-quality matches have lower sourcing cutoffs, so repeated imports raise their share among mature links. This selection generates the gap when $\gamma = 0$. With positive adjustment costs, larger previous-period orders further increase the probability of another positive order.

The third result concerns conditional growth. Let $x_{o,a}^{A,H}$ and $x_{o,a}^{M,H}$ denote orders from an activated and a mature high-quality link at age a , with import values $v_{o,a}^{A,H}$ and $v_{o,a}^{M,H}$. The activated link's first order and import value are $x_{o,0}^U$ and $v_{o,0}^U$.

Proposition 3 (Conditional growth). *Under Proposition 1, compare an activated and a mature high-quality link that differ only in the focal origin's initial order and face the same subsequent states and other-origin orders.*

(i) No adjustment cost. *If $\gamma = 0$ and both links record imports at age $a \geq 1$, then*

$$(\log v_{o,a}^{A,H} - \log v_{o,0}^U) - (\log v_{o,a}^{M,H} - \log v_{o,0}^{M,H}) = \log \frac{v_{o,0}^{M,H}}{v_{o,0}^U} > 0.$$

(ii) Positive adjustment cost. *If $\gamma > 0$ and the positive-order policy is differentiable in x_- with*

$$0 \leq \frac{\partial \log x_o^H(x_-, s)}{\partial \log x_-} < 1,$$

then the same growth gap is strictly positive if both links record imports in every quarter through age a .

Without adjustment costs, a high-quality trial immediately reaches mature scale. With adjustment costs, the elasticity condition makes the initial scale gap close gradually. In both cases, the activated link grows more because it starts smaller.

4 Calibration and model validation

This section calibrates the three models and compares their fit. I first use the same four lifecycle moments to estimate each model, then evaluate the fitted models against additional quarterly moments. I conclude by comparing trial sourcing with a full-information counterfactual.

4.1 Calibration strategy

The quantitative model requires values for β , σ , η , μ , λ_L , λ_H , κ , and γ . I normalize $\lambda_L = 1$, set β , σ , and η externally, and estimate μ , λ_H , κ , and γ internally. The demand scale $\mathcal{D}$ and permanent productivity $\bar{\varphi}$ are absorbed into the scale normalization. The distribution of transitory productivity and the process for delivered import prices are estimated from mature links before the structural calibration.

I set the annual discount factor to 0.96, implying a quarterly value of $\beta = 0.98985$, following [Alessandria, Choi and Ruhl \(2021\)](#). Annual values of 0.95 and 0.97 define the sensitivity range. I set the final-demand elasticity to $\sigma = 3.85$, with sensitivity values of 2.79 and 5.00, following [Antràs, Fort and Tintelnot \(2017\)](#). For the elasticity of substitution across origins, I use 3.24 and 4.12 as alternative values. These are the median TSLS and two-step GMM estimates of the micro Armington elasticity across U.S. goods reported by [Feenstra et al. \(2018\)](#). This micro elasticity is the closest empirical counterpart to substitution across foreign origins in the model's CES input aggregator. Under the normalization $\lambda_L = 1$, λ_H is the match-efficiency ratio relative to the low-quality type.

I jointly estimate μ , λ_H , κ , and γ by matching four lifecycle moments:

$$\hat{m}_C = \begin{bmatrix} \text{activated-mature initial log import-value gap} \\ \text{quarter-one no-import share} \\ \text{four-quarter abandonment} \\ \text{quarter-four conditional import-growth gap} \end{bmatrix}.$$

These moments summarize initial scale, subsequent imports, abandonment, and conditional growth.

For model m , I estimate

$$\hat{\psi}_m = \arg \min_{\psi_m} [\hat{m}_C - m_m(\psi_m)]' W_C [\hat{m}_C - m_m(\psi_m)], \quad W_C = \text{diag}(\hat{\sigma}_j^{-2}).$$

where $\hat{\sigma}_j$ is the firm-clustered standard error of target j . To make model fit comparable, I estimate all three models against the same four data moments using the same inverse-variance weights.

4.2 Calibration fit

Table 1 reports the four data moments and model predictions.

The adjustment-cost model matches initial scale and the quarter-four conditional-growth gap, but it predicts a one-quarter no-import share of 0.102 and four-quarter abandonment of 0.022. Both are close to the mature-link path and far below the data.

The relationship-capital model comes closer on the extensive margin. It predicts a one-quarter no-import share of 0.853 and four-quarter abandonment of 0.527. It also predicts an initial scale ratio of 0.110. However, the accumulation mechanism raises the conditional-growth gap to 0.403 log points, compared with 0.318 in the data.

The trial sourcing model matches all four moments. With four estimated parameters and four targets, the model is just identified.

4.3 Model fit over the sourcing-link lifecycle

Figure 4 evaluates the models away from the four calibration targets. The model parameters are held fixed at the values underlying Table 1.

Panel A reports the share of activated links that record no imports through each age. At age two, the share is 0.682 in the data and 0.700 in the trial sourcing model. At age eight, the corresponding values are 0.473 and 0.389. The adjustment-cost model generates a declining path, but its predictions remain far below the data: 0.043 at age two and 0.016 at age eight. Because match quality is known before sourcing, the first positive order reveals no new information and does not change subsequent sourcing decisions through learning. The relationship-capital model is closer in level but predicts a decline that is too steep, from 0.719 at age two to 0.312 at age eight. New links begin with little relationship capital, which raises early abandonment rates; continued sourcing then builds relationship capital and raises the value of future transactions, causing the predicted no-import share to fall faster than in the data. The trial sourcing model comes closest to both the level and the age profile. Match-quality revelation generates link abandonment after the first positive order. The later decline in abandonment is driven by exogenous productivity shocks.

Panel B reports activated-link import value among links that record imports at each age, relative to mature links. The ratio rises from 0.128 at age one to 0.168 at age eight in the data. The adjustment-cost and trial sourcing models predict 0.116 and 0.192 at these ages. Their paths coincide because they share the same adjustment cost, initial-scale target, and quarter-four conditional-growth target. The relationship-capital model predicts an increase from 0.123 to 0.230, exceeding

the data at later ages. This occurs because accumulating relationship capital raises the probability of survival and import value at the same time. Matching the decline in abandonment rate therefore generates more conditional import-value growth than observed.

Panel C reports activated-link import value relative to mature links, including links that record no imports. This ratio rises from 0.031 at age one to 0.090 at age eight in the data. The adjustment-cost model lies above the data throughout because it predicts that too many activated links record imports after the first positive order, as shown in Panel A. The relationship-capital model rises too quickly because it overpredicts conditional import-value growth, as shown in Panel B. The trial sourcing model most closely matches both the level and the growth of the ratio.

4.4 Information frictions and sourcing margins

I conclude with two exercises comparing trial sourcing with a full-information benchmark that retains the same adjustment costs. The benchmark reveals source–product match quality before activation and otherwise uses the same primitives and parameter vector. The comparison therefore isolates the role of pre-transaction uncertainty.

Extensive and intensive margins. The first exercise asks how pre-transaction uncertainty changes link activation, import value per active link, and total imports. Both regimes face the same distribution of potential source–product matches. Let $r \in \{T, F\}$ denote trial sourcing and full information. Using the order x_{ot}^r and import value v_{ot}^r defined above, I compute

$$\begin{aligned} N_{r,t} &= 100 \Pr(x_{o0}^r > 0, x_{ot}^r > 0), \\ \bar{v}_{r,t} &= \mathbb{E}[v_{ot}^r \mid x_{o0}^r > 0, x_{ot}^r > 0], \\ Y_{r,t} &= 100 \mathbb{E}[\mathbf{1}\{x_{o0}^r > 0\} v_{ot}^r] = N_{r,t} \bar{v}_{r,t}. \end{aligned}$$

Here, $N_{r,t}$ is the number of active initial-cohort links per 100 opportunities, $\bar{v}_{r,t}$ is mean import value per active link, and $Y_{r,t}$ is total import value from the initial cohort. The identity follows because $v_{ot}^r = 0$ whenever $x_{ot}^r = 0$.

Figure 5 reports the three terms. Panel A compares the two regimes along the extensive margin. Full information activates fewer links because firms screen out some low-quality matches before activation. It also avoids the sharp first-quarter decline generated when trial sourcing reveals a low-quality match. The full-information path still declines because some low-quality matches are activated and adverse productivity draws can make their links unprofitable.

Panel B reports import value per active link. Full information conditions activation and the first order on match quality, producing a larger initial order. Trial sourcing starts smaller but expands after the first positive order reveals match quality. Under full information, mean import value rises slightly in quarter one and then declines from 0.278 to about 0.274 as active low-quality links reduce their imports.

Panel C combines these two margins. In the activation quarter, the larger number of trial sourcing links and their smaller first orders nearly offset, leaving total import value 1.2% below full information. From quarter one onward, the extensive-margin gain dominates as the conditional means converge. By quarter four, total import value from the initial cohort under trial sourcing is 24.69, compared with 22.53 under full information, a 9.6 percent increase. Uncertainty before sourcing raises later cohort imports because firms try matches that they would reject under full information.

The paths converge as match quality becomes known in both regimes. Differences persist because information at activation changes both the composition of the cohort and the distribution of previous-period orders.

Selection. The second exercise shows how pre-transaction uncertainty changes the timing of selection. For $z \in \{L, H\}$, define

$$p_{r,t}(z) = \Pr(x_{o0}^r > 0, x_{ot}^r > 0 \mid \lambda_o = \lambda_z), \quad q_{r,t}^H = \frac{\mu p_{r,t}(H)}{\mu p_{r,t}(H) + (1 - \mu)p_{r,t}(L)}.$$

Panel A of Figure 6 reports activation probabilities by match type, Panel B reports the high-type share among active links, and Panels C and D report type-specific activity paths.

In the calibrated trial sourcing model, both high- and low-quality matches are activated with probability one. After the first positive order, low-quality matches are less likely to remain active. Their activity probability falls to 0.901 in quarter one and 0.871 in quarter eight, raising the high-type share among active links from 5.73 to 6.52 percent. Under full information, selection occurs before trading: the activation probability is one for a high-quality match and 0.884 for a low-quality match. Full information therefore shifts selection against low-quality matches from subsequent continuation to initial activation.

5 Conclusion

This paper documents a joint lifecycle of newly activated import links. Within the same importer–product–quarter, an activated origin begins at 10.6 percent of mature-origin scale. Four-quarter abandonment is 57.3 percent among activated links and 2.2 percent among mature links. Among activated links that record imports in quarter four, the activated–mature conditional-growth gap is 0.318 log points.

The trial sourcing model explains this joint pattern through uncertainty about source–product match quality. Match quality is unknown before sourcing and revealed by the first positive order. This uncertainty reduces the first order and shifts selection from activation to continuation. Imports from high-quality activated origins then grow from a smaller base. When the three models are calibrated to the same four moments, the adjustment-cost model generates too little abandonment and the relationship-capital model generates too much conditional growth. The trial sourcing model matches the calibration targets and fits the remaining quarterly lifecycle moments more closely.

The comparison with a full-information counterfactual shows how uncertainty changes sourcing margins. Trial sourcing activates more links but places smaller initial orders. Total import value from the initial cohort is therefore 1.2 percent lower than under full information in the activation quarter but 9.6 percent higher by quarter four. Because activation does not condition on match quality under trial sourcing, low-quality matches account for a larger share of activated links.

When only a positive order reveals match quality, importers activate more links, place smaller first orders, and shift later imports toward links that continue to record imports.

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Figures and Tables

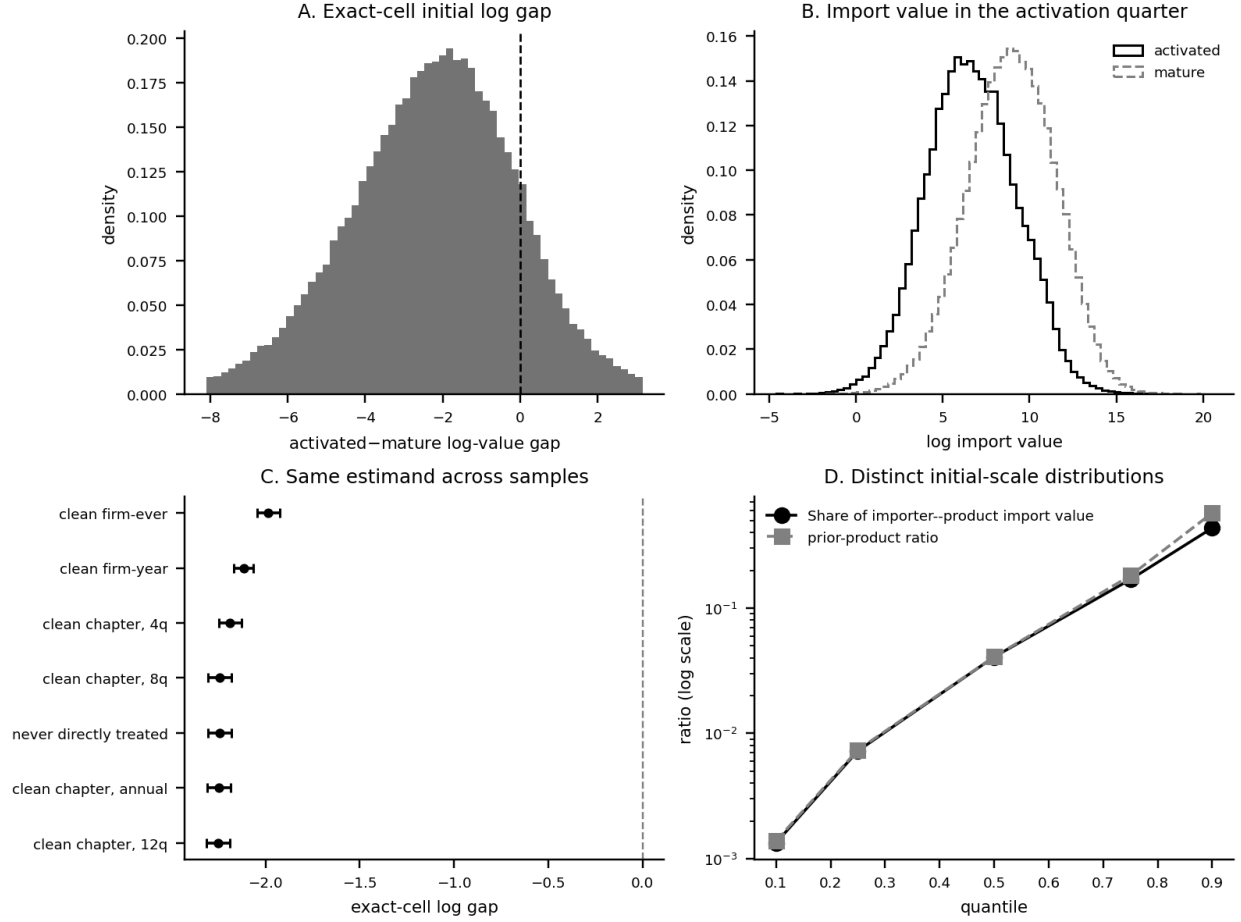


Figure 1: Activated origins start small

Notes: Panel A plots the distribution of exact-cell activated-mature log-value gaps. Panel B compares import-value distributions in the activation quarter. Panel C repeats the activated-mature comparison across product definitions, firm exposure, no-import windows, annual aggregation, and an earlier sample definition. Panel D reports quantiles of the activated origin's share of importer-product import value and its import value relative to pre-activation import value. The main comparison contains 127,975 cells across 7,133 firms.

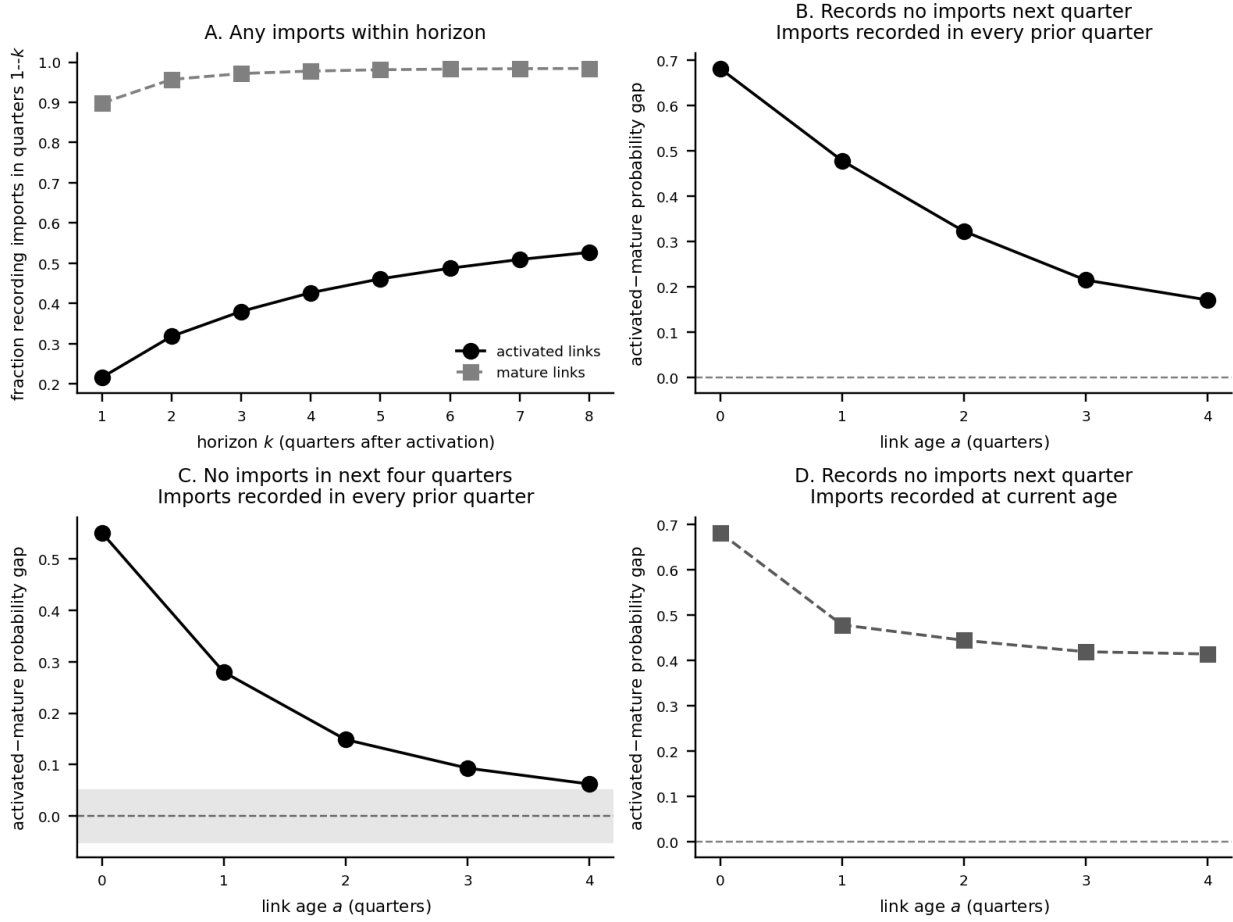


Figure 2: Subsequent imports and abandonment over link age

Notes: Let t denote the activation quarter. Panel A reports the cumulative fraction of links that record imports between $t + 1$ and $t + k$. Panels B–D report activated–mature gaps by link age a , with $a = 0$ in the activation quarter. Panel B conditions on imports in every quarter through age a and reports the probability of no imports in the next quarter. Panel C uses the same risk set and reports the probability of no imports in the next four quarters; at age zero, this outcome is abandonment. Panel D conditions only on imports at age a , allowing a link to return after an earlier quarter with no imports, and reports the probability of no imports in the next quarter. Mature origins are rematched within the importer–product–quarter at each age and must satisfy the maturity criterion then. The shaded region in Panel C is the pre-specified ± 0.05 equivalence band.

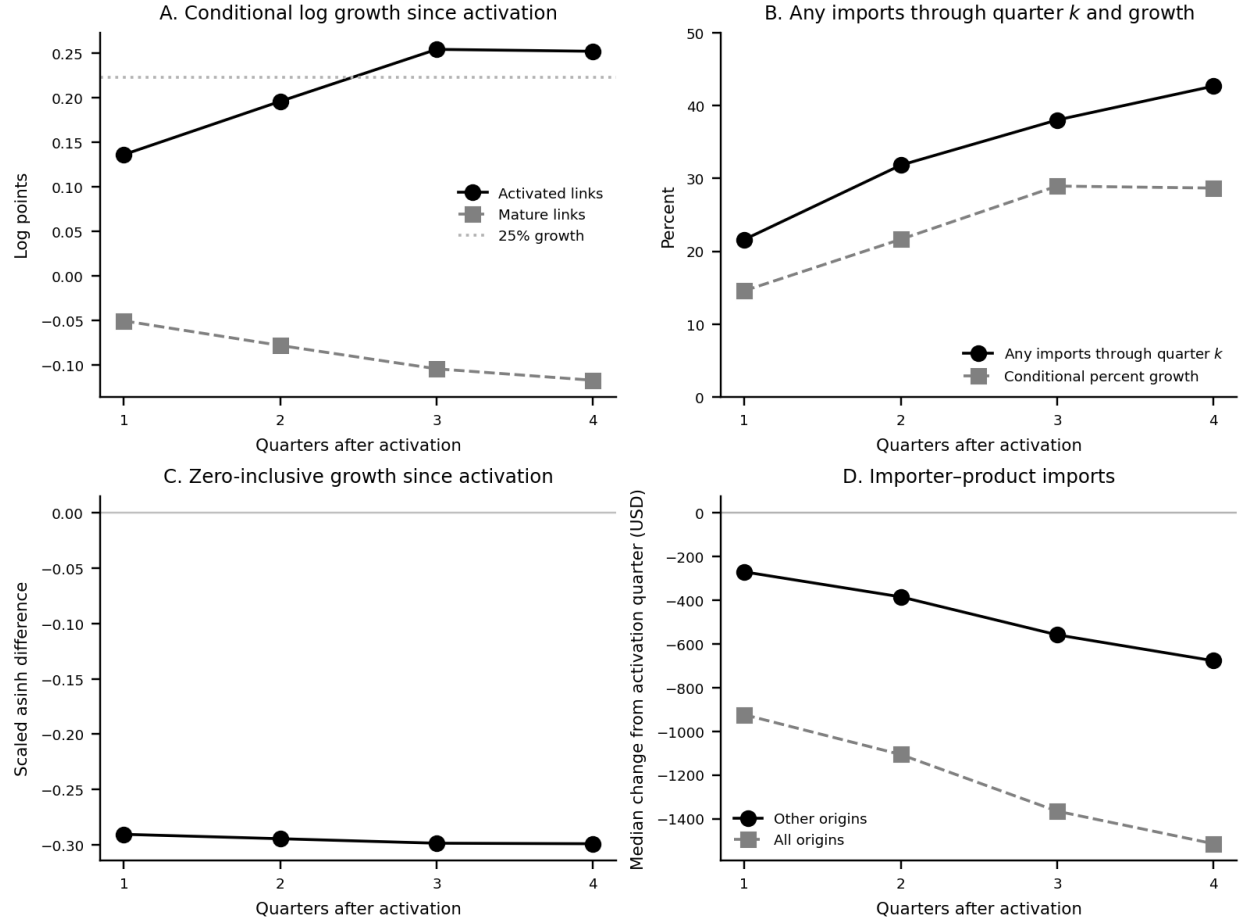


Figure 3: Growth since activation, subsequent imports, and sourcing-portfolio adjustment

Notes: Panel A reports conditional log growth since activation separately for activated and mature links that record imports in quarter k ; the dotted line marks $\log(1.25)$. Panel B reports the fraction of activated links that record imports in any quarter from 1 through k and the conditional percent change obtained by exponentiating mean activated-link log growth in Panel A. Panel C reports zero-inclusive growth since activation, measured by the scaled inverse-hyperbolic-sine difference; links that record no imports in quarter k enter with zero import value. Panel D reports link-level median changes in import value from other origins and from all origins for the same importer-product.

Table 1: Calibration fit

Moment	Data	S.E.	Adjustment costs	Relationship capital	Trial sourcing
Initial activated–mature log import-value gap	−2.242	0.0316	−2.242	−2.204	−2.242
Quarter-one no-import share	0.784	0.0030	0.102	0.853	0.784
Four-quarter abandonment	0.573	0.0038	0.022	0.527	0.573
Quarter-four conditional-growth gap	0.318	0.0201	0.318	0.403	0.318
Standardized RMSE			134.933	13.310	0.000

Notes: Each model is estimated using the same four moments and diagonal inverse-variance weights. Standard errors are clustered by firm. The initial log import-value gap of -2.242 corresponds to an activated–mature import-value ratio of 0.106. The trial sourcing model has four estimated parameters and is therefore just identified. The adjustment-cost model makes match quality known before sourcing; the relationship-capital model adds an accumulating relationship-specific state.

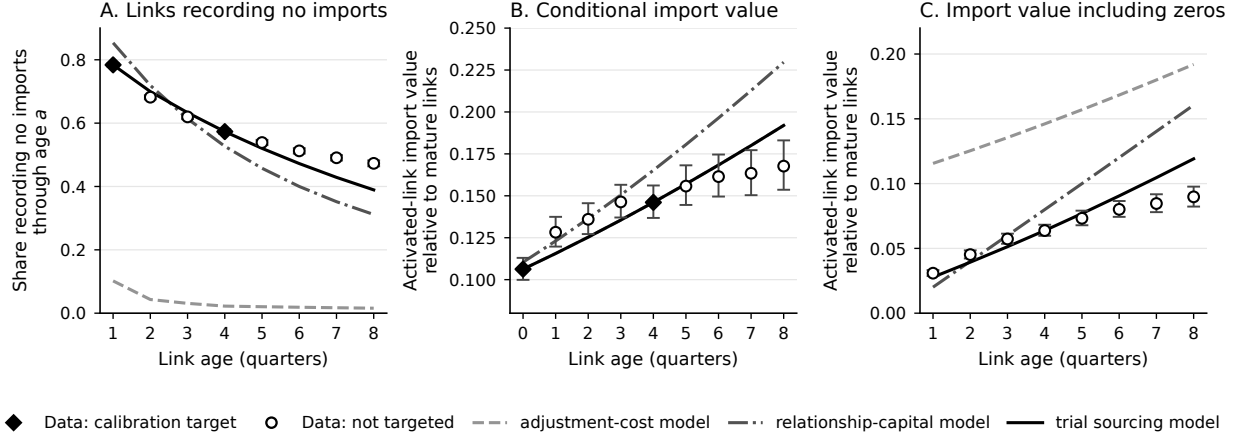


Figure 4: Model fit over the sourcing-link lifecycle

Notes: The figure compares quarterly data with predictions from the adjustment-cost, relationship-capital, and trial sourcing models. Filled diamonds mark calibration targets; open circles mark moments not used in estimation. Vertical bars report pointwise 95 percent confidence intervals clustered by firm; some are narrower than the symbols. Panel A reports the share of activated links that record no imports through each age; the age-four point is four-quarter abandonment. Panel B reports activated-link import value among links that record imports at each age, relative to mature links. The adjustment-cost and trial sourcing paths coincide in Panel B because they share the same adjustment-cost specification, initial-scale target, and quarter-four conditional-growth target. Panel C reports activated-link import value relative to mature links, including links that record no imports.

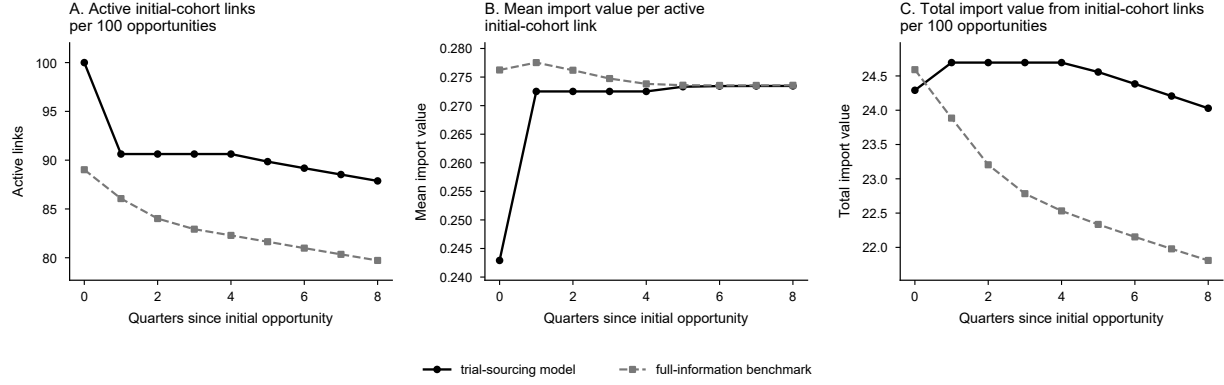


Figure 5: Information frictions and sourcing margins

Notes: The figure compares trial sourcing with the full-information counterfactual for the same distribution of potential source-product matches. Panel A reports $N_{r,t} = 100 \Pr(x_{o0}^r > 0, x_{ot}^r > 0)$. Panel B reports $\bar{v}_{r,t} = \mathbb{E}[v_{ot}^r \mid x_{o0}^r > 0, x_{ot}^r > 0]$. Panel C reports $Y_{r,t} = 100 \mathbb{E}[\mathbf{1}\{x_{o0}^r > 0\} v_{ot}^r] = N_{r,t} \bar{v}_{r,t}$. The initial cohort is fixed at $t = 0$; links may place no order in an intervening period and later resume. Panel C covers only this cohort.

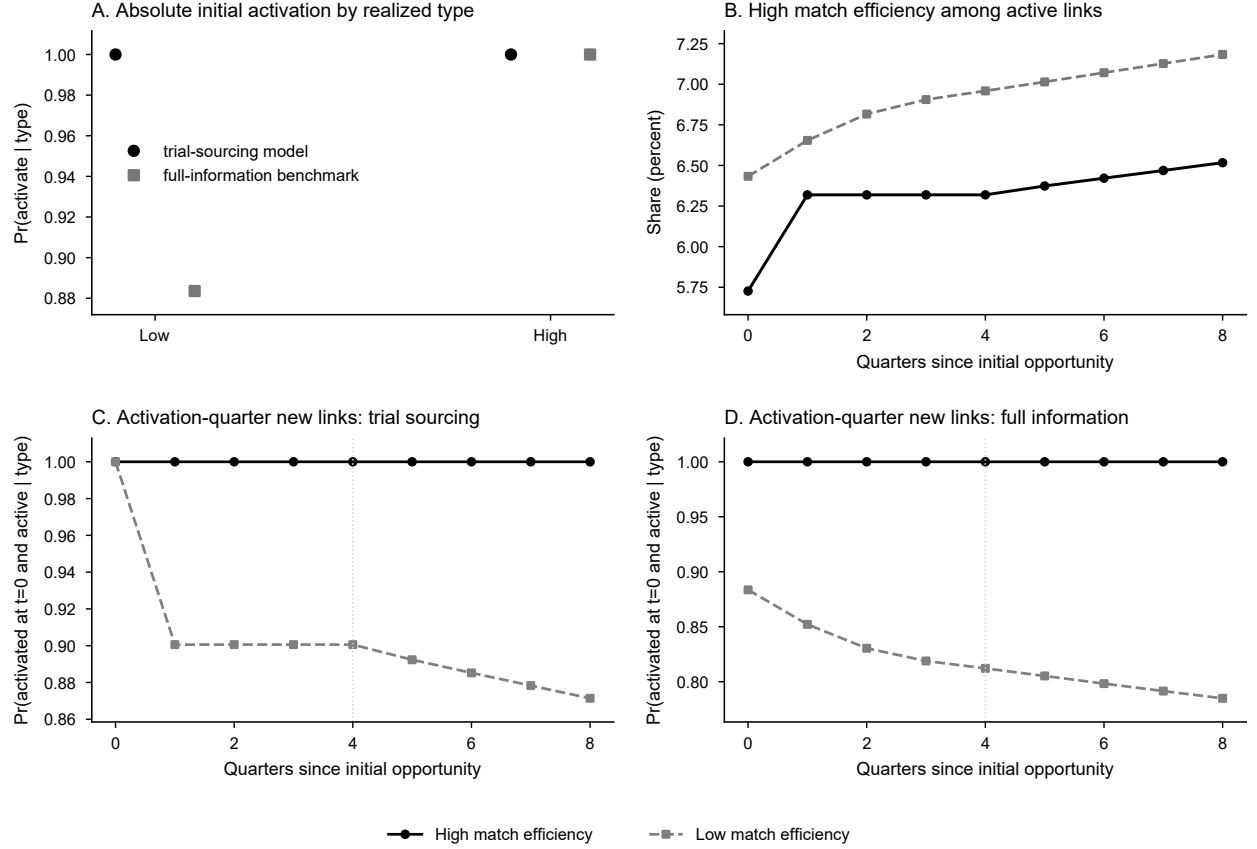


Figure 6: Screening and selection by match quality

Notes: For $z \in \{L, H\}$, let $p_{r,t}(z) = \Pr(x_{o0}^r > 0, x_{ot}^r > 0 \mid \lambda_o = \lambda_z)$. Panel A reports $p_{r,0}(z)$, Panel B reports the high-type share among active initial-cohort links, and Panels C and D report $p_{r,t}(z)$ under trial sourcing and full information. These probabilities are unconditional on activation. All links have a zero previous-period order at $t = 0$. In the calibrated model, every high-quality match receives a positive first order, so its activation probability is one.

A Data Construction and Alternative Sample Definitions

A.1 Data construction and censoring

I aggregate Colombian customs declarations to quarterly importer–product–origin cells. A concordance maps HS2007, HS2012, and HS2017 codes into 5,229 stable product groups and covers 99.1 percent of import value. The harmonized panel contains 8.88 million quarterly observations. After excluding HS2 chapters containing products directly subject to DS461, the baseline sample contains 4,455 products and 8,266,306 observations.

Link activation requires a complete eight-quarter no-import window and prior imports of the same product from another origin. Each outcome sample must be observed through its full horizon; sample-end missingness is never coded as no imports. A reactivation is an activation preceded by earlier imports within the sample.

A.2 Alternative activation and sample definitions

The activated–mature initial-scale gap is stable across product samples, firm restrictions, no-import windows, and annual aggregation. The implied scale ratios range from 0.098 to 0.137.

Table A.1: Initial scale across link-activation and sample definitions

Definition	Log gap	95% lower	95% upper	Scale ratio
Clean chapter, eight-quarter no-import window	−2.242	−2.303	−2.181	0.106
Never directly treated	−2.242	−2.304	−2.180	0.106
Clean firm-year	−2.116	−2.169	−2.063	0.121
Clean firm-ever	−1.985	−2.045	−1.924	0.137
Legacy HS61–63	−2.321	−2.717	−1.924	0.098
Clean chapter, four-quarter no-import window	−2.187	−2.247	−2.127	0.112
Clean chapter, twelve-quarter no-import window	−2.252	−2.313	−2.190	0.105
Clean chapter, annual definition	−2.247	−2.308	−2.186	0.106

Notes: Each row reports the equal-cell mean activated–mature log-value gap. The scale ratio exponentiates this gap. Confidence intervals are clustered by importer. Clean firm-year excludes importer-years with imports of treated products; clean firm-ever excludes importers that ever import a treated product.

B Heterogeneity and Robustness

B.1 Product groups, cohorts, and influence

An HS2 chapter enters the analysis if it contains at least 100 activated–mature cells and 30 firms. Each outcome also requires at least 100 valid cells. Fewer chapters meet the requirement for growth because this outcome conditions on imports four quarters after activation.

The initial-scale gap is negative in all 61 chapters that meet the sample requirement, and the four-quarter abandonment gap is positive in all 60. Among the 28 chapters for which age-four growth can be estimated, imports from activated origins grow in 27; simultaneous 95 percent confidence intervals exclude zero in 14.

All three patterns also hold in each quartile of pre-activation importer–product import value, for first-observed and reactivated links, and across three calendar periods. For first-observed links, the initial log gap is −2.235, the abandonment gap is 57.1 percentage points, and age-four conditional growth is 0.265 log points. The corresponding estimates for reactivated links are −2.259, 49.2 percentage points, and 0.223 log points.

Leave-one-HS2-out estimates retain their signs and at least half of their baseline magnitudes. Removing HS84 produces the largest changes: 0.7 percentage point for abandonment and 0.018 log points for age-four growth. No single HS2 chapter drives the lifecycle facts.

C Model Derivations

C.1 Trial sourcing

C.1.1 Pricing and the minimum order

Inverse demand is $p = (\mathcal{D}/y)^{1/\sigma}$. Substituting $y = \varphi M(\mathbf{x})$ gives Equations (1) and (2); maximizing profit over p gives Equation (3). At $x = 0$, operating surplus net of the fixed order cost is $-\kappa < 0$. By continuity, no sufficiently small positive order covers this cost. Hence $\underline{x}_o > 0$ whenever a positive order is viable. The optimal first order may exceed this lower bound.

C.1.2 Concavity and increasing differences

Holding s and other-origin orders fixed, write $B_{-o} = \sum_{j \neq o} (\lambda_j x_j)^\rho$. For $x > 0$, $p_o^m + \pi_{o,x} > 0$, and direct differentiation gives

$$\begin{aligned} \frac{\partial}{\partial \log \lambda_o} \log(p_o^m + \pi_{o,x}) &= \frac{\rho B_{-o} + \alpha(\lambda_o x)^\rho}{B_{-o} + (\lambda_o x)^\rho} > 0, \\ \frac{x \pi_{o,xx}}{p_o^m + \pi_{o,x}} &= \frac{(\rho - 1)B_{-o} + (\alpha - 1)(\lambda_o x)^\rho}{B_{-o} + (\lambda_o x)^\rho} < 0. \end{aligned}$$

Because $\alpha, \rho \in (0, 1)$, operating surplus is strictly concave in the order and has strict increasing differences in order size and match efficiency.

C.1.3 Initial-scale gap

Proof of Proposition 1. Fix productivity, delivered import prices, and other-origin orders. Because both orders are positive, the fixed order cost does not affect their marginal values. At any common order x , the mature-link marginal objective minus the trial marginal objective is

$$\begin{aligned} (1 - \mu) \Big\{ \pi_{o,x}(x; \lambda_H) - \pi_{o,x}(x; \lambda_L) \\ + \beta \mathbb{E}[V_{o,x-}(x, s'; H) - V_{o,x-}(x, s'; L)] \Big\} + \gamma p_o^m x_- . \end{aligned}$$

Strict increasing differences, $0 < \mu < 1$, and the continuation-value condition make this expression positive. At the interior trial optimum the trial marginal objective is zero. Strict concavity therefore gives $x_o^H(x_-, s) > x_o^U(s)$, and the common delivered import price preserves this ranking. $\square$

C.1.4 Next-quarter no-import gap

Proof of Proposition 2. First set $\gamma = 0$. A revealed type records imports when maximized operating surplus covers κ . Because operating surplus increases in λ_o and φ , $\varphi_{o\tau}^H < \varphi_{o\tau}^L$. For a fixed path of delivered import prices and other-origin orders, let

$$q_{j\tau} = 1 - F_\varepsilon \left(\log \frac{\varphi_{o\tau}^j}{\bar{\varphi}} \right), \quad j \in \{L, H\},$$

be the probability that type j records imports in quarter τ . Positive density between the cutoffs implies $q_{H\tau} > q_{L\tau}$. The maturity criterion therefore changes the high-to-low odds from $\mu/(1 - \mu)$ to

$$\frac{\mu}{1 - \mu} \prod_{\tau=t-4}^t \frac{q_{H\tau}}{q_{L\tau}} > \frac{\mu}{1 - \mu}.$$

By contrast, activation is independent of the unobserved type, so the high-quality share at activation remains μ . Because $q_{H,t+1} > q_{L,t+1}$, the cohort's next-quarter no-import probability decreases strictly with its high-quality share. The stated inequality follows.

Now let $\gamma > 0$. Relative to a zero order, the adjustment-cost contribution of a positive order is

$$-\Psi_o(x, x_-; p_o^m) + \Psi_o(0, x_-; p_o^m) = -\frac{\gamma p_o^m}{2} x^2 + \gamma p_o^m x x_-,$$

which increases in x_- . Under the stated monotonicity conditions, repeated imports select mature links toward high-quality matches, and a larger current order weakly raises next-quarter participation. The mature cohort therefore has both a larger high-quality share and, type by type, a weakly larger previous-period order than the activated cohort. The strict type difference gives the stated no-import gap. $\square$

C.1.5 Conditional import-value growth

Proof of Proposition 3. Proposition 1 gives $v_{o,0}^U < v_{o,0}^{M,H}$ at a common delivered import price.

If $\gamma = 0$, the previous-period order drops from the problem. Conditional on a high-quality match and a common age- a state, $v_{o,a}^{A,H} = v_{o,a}^{M,H}$. Hence

$$\begin{aligned} & (\log v_{o,a}^{A,H} - \log v_{o,0}^U) - (\log v_{o,a}^{M,H} - \log v_{o,0}^{M,H}) \\ &= \log \frac{v_{o,0}^{M,H}}{v_{o,0}^U} > 0. \end{aligned}$$

For $\gamma > 0$, the mean-value theorem and the elasticity restriction imply that each positive log-order gap lies between zero and the preceding gap, with a strict upper inequality whenever the preceding gap is positive. Iterating along the common state sequence and using the common delivered import price gives

$$\begin{aligned} 0 &\leq \log \frac{v_{o,a}^{M,H}}{v_{o,a}^{A,H}} < \log \frac{v_{o,0}^{M,H}}{v_{o,0}^U}, \\ & (\log v_{o,a}^{A,H} - \log v_{o,0}^U) - (\log v_{o,a}^{M,H} - \log v_{o,0}^{M,H}) \\ &= \log \frac{v_{o,0}^{M,H}}{v_{o,0}^U} - \log \frac{v_{o,a}^{M,H}}{v_{o,a}^{A,H}} > 0. \end{aligned}$$

$\square$